placement among the other boats in the race. So instead, Amodei took the easy way and told the boat to maximise its score. He assumed that the score was a good proxy for winning the race. After all, it worked with the car races.

But the boat race had a peculiar feature, absent from the car races, that allowed the ingenious AI to find a loophole in the game's rules. The game rewarded players with a lot of points for getting ahead of other boats – as in the car races – but it also rewarded them with a few points whenever they replenished their power by docking into a harbour. The AI discovered that if instead of trying to outsail the other boats, it simply went in circles in and out of the harbour, it could accumulate more points far faster. Apparently, none of the game's human developers – nor Dario Amodei – had noticed this loophole. The AI was doing exactly what the game was rewarding it to do – even though it is not what the humans were hoping for. That's the essence of the alignment problem: rewarding A while hoping for B.[27] If we want computers to maximise social benefits, it's a bad idea to reward them for maximising user engagement.

A third reason to worry about the alignment problem of computers is that because they are so different from us, when we make the mistake of giving them a misaligned goal, they are less likely to notice it or request clarification. If the boat-race AI had been a human gamer, it would have realised that the loophole it found in the game's rules probably doesn't really count as 'winning'. If the paper-clip AI had been a human bureaucrat, it would have realised that destroying humanity in order to produce paper clips is probably not what was intended. But since computers aren't humans, we cannot rely on them to notice and flag possible misalignments. In the 2010s the YouTube and Facebook management teams were bombarded with warnings from their human employees – as well as from outside observers – about the harm being done by the algorithms, but the algorithms themselves never raised the alarm.[28]

As we give algorithms greater and greater power over health care, education, law enforcement and numerous other fields, the alignment problem will loom ever larger. If we don't find ways to solve it, the consequences will be far worse than algorithms racking up points by sailing boats in circles.